

Excel to MARC Automated Processing Pipeline for Archival Metadata

This script is a Python pipeline designed for **Google Colab** that processes historical/archival document metadata from an Excel file (specifically related to the *Francke Nachlass* collection) and converts it into standard **MARC21 format (.mrk)** for library and archival systems.

Key Functions and Execution Workflow:

1. Cell 1: Environment Setup & File Handling

- **Dependencies:** Installs openpyxl (for Excel files) and beautifulsoup4 (for HTML parsing).
- **Interactive Upload:** Prompts the user to upload two Excel files via Google Colab (files.upload()):
 - **Main Metadata File:** Excel file containing document properties (title, author, dates, notes, URLs, etc.).
 - **Location Mapping File:** sender_place_new.xlsx, used to normalize and map sender location names.
- **Data Loading & Inspection:** Reads both spreadsheets into Pandas DataFrames and displays basic statistics and preview samples.

2. Cell 2: Core Conversion Engine (Excel to MARC21 .mrk)

- **System Setup & Data Cleaning:** Sets up logging structures (MARC_MASTER_LOG.csv, MARC_LOG_SUMMARY.csv) and generates run timestamps for MARC control fields (005 and 008). Defines helper functions (clean, normalize_space, strip_html) to strip formatting, resolve missing values (NaN/None), and handle special characters using unicode normalization.
- **Authority Record Link Parsing & Entity Grouping:**
 - **parse_authority_links & group_authorities:** Parses linked HTML anchors containing authority database URLs (Library of Congress, VIAF, Wikidata, GeoNames).

- Groups multiple links corresponding to the same entity based on URI domain types and string matching, prioritizing authority sources in the hierarchy: **LC > VIAF > Wikidata > GeoNames**.
- **Sender Place Normalization:**
 - Cross-references raw sender location strings with sender_place_new.xlsx to replace local or historical place names with standardized geographical forms.
- **MARC21 Record Builder (convert_row):** Iterates through each row in the Excel dataset to build standard MARC tags:
 - **Control Fields:** Generates LDR (Leader), 001 (Control number placeholder), 005 (Date/Time modified), 008 (Fixed-length data including date of record creation, language code, and year).
 - **Identifiers & System Numbers:** Maps 024 (Other Standard Identifier extracted from custom citations), 035 (System Control Number using repository and identifier), 040 (Cataloging Source), 041 (Language Code), 043 (Geographic Area Code), 045 (Time Period of Heading), 046 (Special Coded Dates in YYYYMMDD format), and 049 (Local Holdings).
 - **Creators & Titles:**
 - 100: Primary personal name creator (applies authority URI indicators \$0 for LC, \$1 for VIAF/Wikidata).
 - 245: Main Title Statement, appending \$c statement of responsibility derived from primary creator.
 - 246: Variant/Alternative Title.
 - **Publication/Distribution & Physical Description:**
 - 264: Production/Publication statement (Sender place and date).
 - 300: Physical Description (e.g., 1 online resource (letter)).
 - 336, 337, 338: RDA Content, Media, and Carrier Type codes for digital still images and documents.

- **Notes:** Maps 500 (Location notes), 505 (Transcriptions/Contents), 506 (Restrictions on access/embargo date), 520 (Abstract), 524 (Preferred Citation), 530 (Additional Physical Form Available), and 540 (Terms Governing Use and Reproduction/Licenses).
- **Subject Access Fields:**
 - 600: Personal name or family subject headings with authority URIs.
 - 610: Corporate body subject headings or recipient organization matching.
 - 650: Topical subject headings (LCSH and FAST terms with \$2fast).
 - 651: Geographic subject headings with GeoNames/Wikidata URIs (\$2geonames, \$2wikidata).
 - 653: Uncontrolled index term keywords.
 - 655: Index term for Genre/Form (\$2lcgft).
- **Logging:** Records every field generation step, value substitution, or skipped field into a log for auditability.

Project Team

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